

Skills Summary

Finding Inverses with Domain Restrictions

Use this reference while completing your worksheet or when studying.

When an inverse exists. f has an inverse function exactly when f is **one-to-one** (each output comes from one input — the horizontal line test). If it is not, restrict the domain to a piece on which it is, and say which piece you kept.

The swap that never changes: domain of f^{-1} = range of f , and range of f^{-1} = (restricted) domain of f . An inverse written without its domain is only half an answer.

The procedure: write $y = f(x)$, swap x and y , solve for y , then choose the branch or sign that the restriction forces.

1. Invert a one-to-one function and state its domain

Linear, radical and simple rational functions are already one-to-one, so no restriction is needed — but the domain still has to be read off the *range* of f . Undo the operations in reverse order: the last thing f does is the first thing f^{-1} undoes.

Example: $f(x) = 2x + 9$. Find f^{-1} and then $f^{-1}(15)$.

Step 1. A non-constant linear function is one-to-one on all of the reals, so there is nothing to restrict — go straight to the swap.

Step 2. $y = 2x + 9$; swap and solve: $x = 2y + 9$, so $f^{-1}(x) = \frac{x-9}{2}$, defined for every real x . Then $f^{-1}(15) = \frac{15-9}{2}$.
 $\Rightarrow f^{-1}(15) = 3$

Try it: $f(x) = 5x - 2$. Find f^{-1} , then compute $f^{-1}(18)$.

check: 4

2. Restrict a quadratic, then pick the branch

A parabola fails the horizontal line test, so restrict at the **vertex**: $x \geq h$ keeps the right branch, $x \leq h$ the left one. Complete the square to find h .

Solving for y produces $\pm\sqrt{}$. The kept branch chooses the sign: the outputs of f^{-1} must lie in the restricted domain of f , so the right branch takes $+$ and the left branch takes $-$.

Example: $f(x) = (x-1)^2$ restricted to $x \geq 1$. Find f^{-1} and $f^{-1}(9)$.

Step 1. The vertex sits at $x = 1$ and the restriction keeps the right branch, so the inverse's outputs must all be at least 1 — that will fix the sign.

Step 2. $y = (x-1)^2$; swap and solve: $y = 1 \pm \sqrt{x}$, and the plus sign is the one with outputs above 1. So $f^{-1}(x) = 1 + \sqrt{x}$ with domain $x \geq 0$, and $f^{-1}(9) = 1 + 3$.

$\Rightarrow f^{-1}(9) = 4$

Try it: $f(x) = (x + 3)^2$ restricted to $x \geq -3$. Compute $f^{-1}(25)$.

check:

3. Verify an inverse pair by composition

Two functions are inverses only if *both* compositions collapse: $f(g(x)) = x$ and $g(f(x)) = x$, each on the appropriate domain. One direction alone is not enough when a domain has been restricted.

A single test value is a fast sanity check; the algebraic composition is the proof.

Example: Show $g(x) = \frac{x+4}{3}$ is the inverse of $f(x) = 3x - 4$, then compute $g(14)$.

Step 1. Composition is the definition, so substitute f into g and simplify rather than re-deriving anything.

Step 2. $g(f(x)) = \frac{(3x-4)+4}{3} = \frac{3x}{3} = x$, so the pair is inverse. Then $g(14) = \frac{14+4}{3}$.
 $\Rightarrow g(14) = 6$

Try it: Check that $g(x) = \frac{x+6}{2}$ inverts $f(x) = 2x - 6$, then compute $g(4)$.

check:

(!) Watch out: f^{-1} never means $\frac{1}{f}$. And an inverse formula without its domain is a different function: $x^2 + 4$ is defined everywhere, but the inverse of $\sqrt{x-4}$ is $x^2 + 4$ *only for* $x \geq 0$.